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Combining Graph Neural Network with Attention Mechanism in Sequence-Based Music Recommendation System

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***Abstract*—Music recommendations have become an important part of everyday life. With the abundance of music available, it is becoming increasingly difficult to find suitable music content. To address this issue, recommendation systems have emerged as a solution to personalize song suggestions for users. Traditional recommendation systems often fail to capture sequential patterns of user interaction. To understand these patterns, recommendation systems are based on a sequential approach. However, traditional sequential recommendation system models do not sufficiently capture dependencies on long-term temporal patterns. In this study, we apply a combination of Graph Neural Network (GNN) with attention mechanisms to the Music4All dataset to address sequential music recommendations. This approach, which adapts the Graph-based Attentive Sequential model with Metadata (GASM), This model was applied for the first time to this music dataset, which has a wider diversity and distribution of users and can provide new insights into the effectiveness of this model. GNN capture the relationships between users and items, as well as between items themselves, while the attention mechanism captures user interest patterns over time within a session. We also incorporate long-term and dynamic preferences to enrich the context of user preferences. By applying this model to the Music4All dataset, the model can recommend the next music track with a Hit Ratio (HR) evaluation ranging from 32.69 to 49.16 and a Mean Reciprocal Rank (MRR) ranging from 23.34 to 30.71. These results show that our approach can effectively capture sequential patterns in music in large-scale datasets.**

***Index Terms*—Sequential recommendation system, music, graph neural network, attention mechanism, preference**

I. Introduction

In this digital age, music has become almost inseparable from everyday life. This is followed by the growing music production. The development of music production makes the number of songs available to consumers also become more and more [1]. This makes it more difficult for consumers to find music content that suits their preferences because they will spend a lot of time searching for music that suits their tastes, which can reduce the user experience in finding music [2]. To reduce the burden on consumers in finding music content, recommendation systems play an important role in this regard.

A recommendation system is a system that can personalize with the aim of providing product suggestions that may be of interest to users [3]. The main goal is to improve the user experience. Improved user experience in recommendation systems can be achieved from user interactions over time. In this case the recommendation can be based on a time sequence approach called a sequential recommendation system.

Sequential Recommendation System (SRS) are tasked with analyzing user interaction history to predict consumer preferences [4]. User interactions can be a history of accessing sites, listening to songs, watching videos, etc. The interaction history will usually form a time sequence pattern following the transition from one item to the next. Using interaction time sequence patterns, the recommendation system will accurately understand user preferences. One of the implementations of sequential recommendation systems is through traditional approaches, such as markov models that provide item predictions by considering the last interaction. However, such models cannot handle long-term sequence dependency [5]. In addition, there are latent factor models that work by learning latent representations to analyze transitions between items. However, transition models often vary across different data, which makes them in a position to capture complex transition relationships [5].

To overcome sequence dependency, researchers have conducted several experiments, one of which uses a neural network graph representing user interactions with items and attention mechanisms that provide important weights in the order of user history. Research by Chang et al. [6], preferences in short-term transitions has integrated a Graph Neural Network (GNN) model as a graph of user relationships with items and attention mechanisms as weights in assessing. The integration of the model showed better results than the baseline in handling the dependence of time sequence [6].

In this study, we adapt the Graph-based Attentive Sequential model with Metadata (GASM) approach [7]. The model works by combining a GNN with an attention mechanism, where the GNN forms connections between users and items, as well as between items and items, while the attention mechanism forms short-term preferences by weighting items based on changes that occur over time during a session. In addition, other preferences are built dynamically over the long term, creating complex preferences.

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The main contribution of this study lies in the application of a recommendation model that has been adapted to the different data characteristics from previous studies. The previous GASM models used datasets with relatively limited numbers of users, such as LastFm with 896 users, 30Music with 2,965 users, and Xiami with 3,982 users, while this study used the Music4All dataset with a much larger number of users, namely 7,998 users. This significant difference allows the model to be tested on a scale that is more representative of real-world use cases. This approach distinguishes our research from the GASM model, which was previously tested on data with a different scope. Thus, this study provides a new perspective on music recommendation modeling based on more diverse user preferences.

II. Related Work

A. Sequential Recommendation System

SRS is a recommendation system with the aim of processing information from a sequence of interactions that are useful in predicting the next item [8]. This system has proven to be very effective in suggesting items or products to users because user interactions with items or products usually occur successively in a time sequence. In contrast to user-based, item-based, and content-based recommendation systems that cannot properly capture user interaction patterns [9]. Some research on sequential recommendation has been done, for example research by Fan et al. [10], uses a stochastic model as a representation of items and self attention to capture the relationship between items in sequence. Another example is research from by Zhang et al. [11], uses a deep learning model with a dynamically formed GNN to model the sequence of user behavior dynamically and link prediction to predict the next item.

B. Graph Neural Network

GNN is a model of deep learning that works on graphs. GNN processes information sets from its neighbor relationships with a message-passing mechanism [12]. In SRS, GNN works to form interaction patterns between users and items. Several research methods regarding GNN in SRS have been conducted with superior results compared to traditional methods. For example, research by Chang et al. [6], showed that the GNN model achieved the highest performance in SRS with a Mean Reciprocal Rank (MRR) value of 0.4 to 0.9.

C. Attention Mechanism

Attention mechanism is a mechanism for the selection process by giving weight to existing inputs [13]. The weight given is formed like a selective focus, where the weight will be grouped at different levels of importance [14]. In SRS, the attention mechanism works to shape user preferences by capturing interaction changes over time. Several research methods regarding attention mechanism in SRS have been carried out, such as the example of research by Kang and McAuley [15], giving satisfactory results by outperforming the baseline on the improvement Hit Ratio (HR) value around 5% to 21% .

III. System Design

This section will explain how the model was developed in this study. The model process is illustrated in Fig. 1, which shows the overall flow of the research design of the combining GNN with attention Mechanism in sequence-based music recommendation system model. Meanwhile, the core design of the model, namely graph construction to preference building, is illustrated in Fig. 2.

A. Dataset

This research uses the Music4All data which is obtained directly from the Google Sites Music4All site: A New Music Database [16]. The Music4All data has many dataset and those used in this research are the id_information and listening_history sections. In the id_information, music metadata can be formed consisting of albums and artists. Meanwhile, listening_history will form information about users with item access to music items. We chose this dataset because it has a large and complex amount of user data and music information, allowing the model to learn many patterns of user history and preferences.

B. Data Prepocessing

Data preprocessing is done to process raw data into ready to use data for SRS. In data preprocessing, we take the first step by forming input data that provides metadata (artist and album) and user information about song listening history. Next, we select the data to compact the data and keep the computing resources stable. In this case, the top 8000 users are selected by ignoring the frequency of music appearing less than 10. Then, we perform index mapping between users and music items with album and artist information. Next, from the item index, time-based sequential data is formed. The formation was done by setting the listening length in the range of 200-500 to maintain the consistency of data from the top users. Finally, we divide the data into 80% train data which contains the previous listening history and 20% test data which contains the latest listening history. From data preprocessing, it produces user

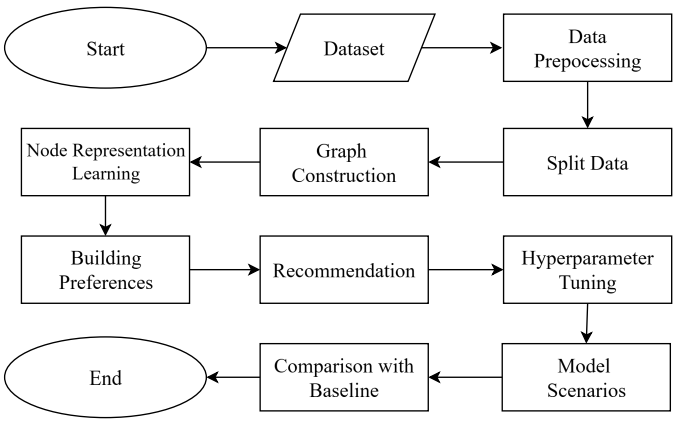

Fig. 1. Research design on combining a GNN with an attention mechanism in a sequence-based music recommendation system.

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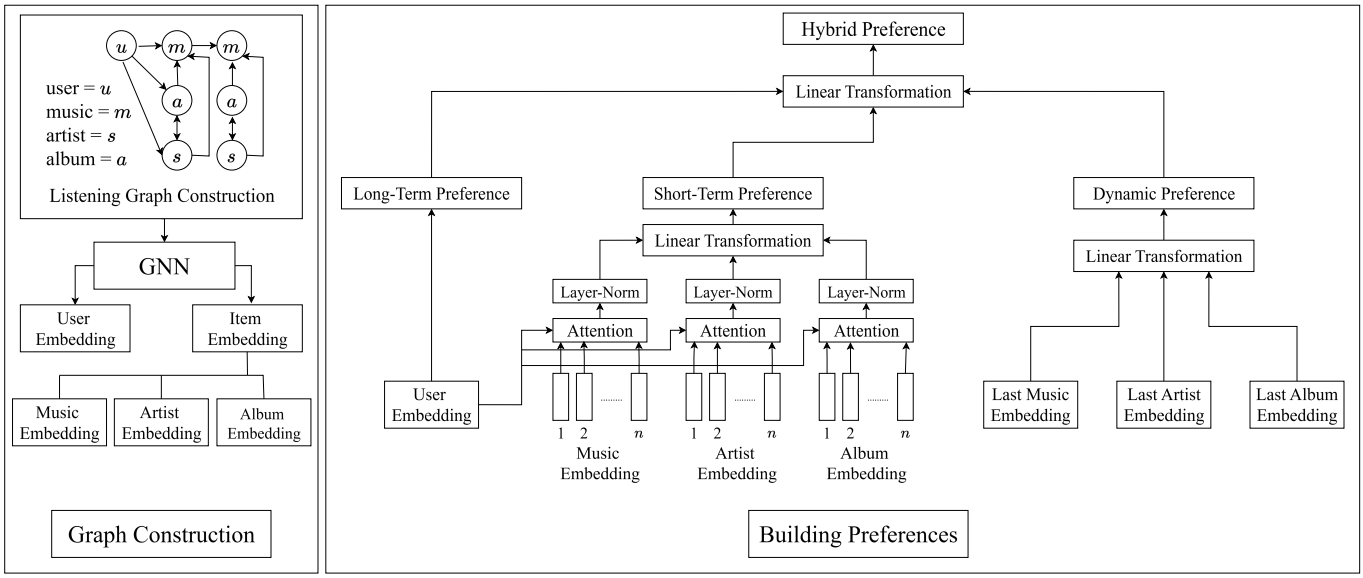

Fig. 2. Illustration of the workflow of graph construction and preference building.

statistics of 7998, music of 56047, artist of 9348, album of 21478, and listening history of 3123667.

C. Graph Construction

User interactions with items are organized in a Listening Graph Construction as shown in Fig. 2. The types of relationships in the graph are u-m (user listens to music), u-a (user accesses album), u-s (user accesses artist information), m-m (sequential relationship between songs), a-m (album contains music), s-m (artist has music), and a-s (bidirectional relationship indicating a strong connection between music information).

D. Node Representation Learning

The construction of latent feature vectors from user nodes and item nodes is performed through a gated GNN, which will produce embeddings as shown in the GNN section in Fig. 2. This construction follows the following formula [7]:

$$\mathbf{a}_{s,i}^t = \mathbf{A}_{s,i}^u : [\mathbf{v}_1^{t-1}, \ldots, \mathbf{v}_n^{t-1}]^\top \mathbf{H} + \mathbf{b} \tag{1}$$

$$\mathbf{z}_{s,i}^t = \sigma(\mathbf{W}_z \mathbf{a}_{s,i}^t + \mathbf{U}_z \mathbf{v}_i^{t-1}) \tag{2}$$

$$\mathbf{r}_{s,i}^t = \sigma(\mathbf{W}_r \mathbf{a}_{s,i}^t + \mathbf{U}_r \mathbf{v}_i^{t-1}) \tag{3}$$

$$\tilde{\mathbf{v}}_i^t = \tanh(\mathbf{W}_o \mathbf{a}_{s,i}^t + \mathbf{U}_o(\mathbf{r}_{s,i}^t \odot \mathbf{v}_i^{t-1})) \tag{4}$$

$$\mathbf{v}_i^t = (1 - \mathbf{z}_{s,i}^t) \odot \mathbf{v}_i^{t-1} + \mathbf{z}_{s,i}^t \odot \tilde{\mathbf{v}}_i^t \tag{5}$$

Equation (1) is performed to obtain the latent feature vector by aggregating the information through the node information ($\mathbf{v}^{t-1}$) to obtain the activation vector ($\mathbf{a}_{s,i}^t$). The activation vector ($\mathbf{a}_{s,i}^t$) will then be used to form an update gate ($\mathbf{z}_{s,i}^t$) in (2) and a reset gate ($\mathbf{r}_{s,i}^t$) in (3) to decide whether the existing information is worth retaining or not. Next, a candidate vector ($\tilde{\mathbf{v}}_i^t$) will be calculated by combining the aggregate information ($\mathbf{a}_{s,i}^t$) and the old vector ($\mathbf{v}_i^{t-1}$) that has been selected by the reset gate ($\mathbf{r}_{s,i}^t$), as in (4). From the calculation of (1) to (4), the latent feature vector ($\mathbf{v}_i^t$) can be generated by combining the old vector ($\mathbf{v}_i^{t-1}$) with the candidate vector ($\tilde{\mathbf{v}}_i^t$) conditioned by the update gate ($\mathbf{z}_{s,i}^t$), as in (5).

E. Building Preferences

We build user preferences into long-term, short-term, and dynamic preferences [7]. The preferences are set as shown in the Building Preferences section in Fig. 2. Long-term preferences reflect users' overall musical tastes, while short-term preferences reflect changes in taste within a single session. Dynamic preferences utilize the last song to predict subsequent listening behavior.

1) Long-Term Preference: The user vector representation shows all the user's history. Thus, this preference can be formulated in (6) [7].

$$\mathbf{P}_u^l = \mathbf{v}_u. \tag{6}$$

Where:

- $\mathbf{P}_u^l$: Long-term preference
- $\mathbf{v}_u$: Embedding of user

2) Short-Term Preference: Changes in user preferences between songs are recorded through an attention mechanism that uses song, album, and artist information from the user's last session with a sequence length of 3 songs, where n is the last song. We split them separately with the notation $x \in \{m, a, s\}$ where m represents music, a represents album, and s represents artist. The short-term preference calculation is formulated as follows [7]:

$$\mathbf{h}_{x,i}^s = \phi(\mathbf{W}_{x,1}^s \mathbf{v}_{x,i} + \mathbf{W}_{x,2}^s \mathbf{v}_{x,n} + \mathbf{b}_x^s) \tag{7}$$

$$\alpha_{u,i,x}^s = \frac{\exp(\mathbf{v}_u^\top \mathbf{h}_{x,i}^s)}{\sum_{i \in U} \exp(\mathbf{v}_u^\top \mathbf{h}_{x,i}^s)} \tag{8}$$

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$$\mathbf{p}_{u,x}^{s} = \sum_{i \in U} \alpha_{u,i,x}^{s} \mathbf{v}_{x,i} \tag{9}$$

$$\mathbf{p}_{u,x}^{s} = \text{layernorm}(\mathbf{p}_{u,x}^{s}) \tag{10}$$

The initial step in building short-term preference is reflected in (7) which represents the context of item x by considering the last item in that session. Next, the attention weight of each item x is calculated as in (8). Equation (9) forms the short-term preference of each item x from the summation of the attention weight with the embedding of item x. Then, the short-term preference of each item x is normalized for model training stability as in (10). To obtain all the short-term preference representations of the three items x, we combine them with a linear transfomation matrix as in (11) [7].

$$\mathbf{p}_{u}^{s} = \mathbf{W}_1[\mathbf{p}_{u,m}^{s} : \mathbf{p}_{u,a}^{s} : \mathbf{p}_{u,s}^{s}] \tag{11}$$

Where:

- $\mathbf{p}_u^s$: Short-term Preference
- $\mathbf{W}_1$: Linear transformation matrix
- $\mathbf{p}_{u,m}^s$: Short-term preference for music
- $\mathbf{p}_{u,a}^s$: Short-term preference for album
- $\mathbf{p}_{u,s}^s$: Short-term preference for artist

3) Dynamic Preference: The vector representation of the music, album, artist of the last music in the user record combined by the linear transformation matrix represents the dynamic preference. Thus, these preferences can be formulated in (12) [7].

$$\mathbf{P}_{u}^{d} = \mathbf{W}_2[\mathbf{v}_{m,n} : \mathbf{v}_{a,n} : \mathbf{v}_{s,n}] \tag{12}$$

Where:

- $\mathbf{P}_u^d$: Dynamic preference
- $\mathbf{W}_2$: Linear transformation matrix
- $\mathbf{v}_{m,n}$: Embedding of music in the last music track
- $\mathbf{v}_{a,n}$: Embedding of the album in the last music track
- $\mathbf{v}_{s,n}$: Embedding of the artist in the last music track

To capture complex preferences, we combine long-term, short-term, and dynamic preferences using a linear transformation matrix as in (13) [7].

$$\mathbf{p}_{u} = \mathbf{W}_3[\mathbf{p}_{u}^{l} : \mathbf{p}_{u}^{s} : \mathbf{p}_{u}^{d}] \tag{13}$$

Where:

- $\mathbf{p}_u$: Hybrid user preference
- $\mathbf{W}_3$: Linear transformation matrix

F. Recommendation

In getting the recommendation score, we construct the probability of the next music $\hat{y}_i$ through multiplying the transponse of the hybrid preference $\mathbf{p}_u^\top$ with the embedding of all items $\mathbf{v}_i$, as in (14) [17].

$$\hat{y}_i = \text{softmax}(\mathbf{p}_u^\top \mathbf{v}_i) \tag{14}$$

For training, we calculate the loss function used to update model parameters through back-propagation by calculating the gradient and passing it backward through the network to optimize weights and improve prediction accuracy. The loss is determined using the categorical cross-entropy formula, where $\mathbf{y}$ is the one-hot encoded vector of the ground truth music, as shown in (15) [18].

$$L(\hat{y}) = -\sum_{i=1}^{|V|} (y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \tag{15}$$

To assess the performance of the model, appropriate evaluation metrics are required. The evaluation metrics will be calculated according to top-k-music. In this case, we use the HR evaluation metric to measure the prediction of the proportion of appropriate music recommendations from real user preferences by dividing the number of music in the hit recommendation list by the number of consecutive tests in the data N, as in (16).

$$\text{HR@k} = \frac{\text{hit}}{N} \tag{16}$$

In addition, we measure the effectiveness of the model in returning relevant items in the ranking to recommend with the MRR as calculated in (17) [18].

$$\text{MRR@k} = \frac{1}{N} \sum_{v_{\text{target}} \in S_{\text{test}}} \frac{1}{\text{Rank}(v_{\text{target}})} \tag{17}$$

Where v_{target} is the target music and $\text{Rank}(v_{\text{target}})$ is the position of the target music in the recommendation list.

In returning the next music recommendation to the user, we divide it into next-one and next-new recommendations [7]. Next-one will return the next music even if the next music has been listened to before. Whereas next-new will only return the next music from new music that the user has never heard before.

G. Hyperparameter Tuning

Model performance is influenced by parameters, so we conducted parameter testing to find the optimal values. The parameters tested were lr with values of 0.005, 0.001, and 0.0005. The parameter l2 was tested with values of 0.001, 0.0001, and 0.00001. The hidden_size parameter was tested with values of 60, 80, and 100. The lr parameter determines the learning rate. Too low a value causes slow learning and suboptimal results, while too high a value causes instability. The l2 parameter determines the regularization value by adding a penalty to the weights to prevent overfitting. The hidden_size parameter determines the dimension of the vector representation so that the model has the capacity to capture data information.

H. Model Scenarios

The model architecture uses three preference components (long-term, short-term, and dynamic). We tested four scenarios to analyze the influence of each component on the recommendation results, namely Full Preferences uses all three

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preference components, W/O Dynamic ignores recent music history, W/O Long-term ignores overall listening history, and W/O Short-term ignores changes in user preferences during the session.

I. Comparison with Baseline

Our model is adapted from the Graph-based Attentive Sequential model with Metadata (GASM), which is an extension of Session-based Recommendation with Graph Neural Networks (SR-GNN) that recommends music based on sessions using GNN embeddings [19]. The main difference is that SR-GNN uses a homogeneous graph without utilizing the entire item history, while our model constructs a heterogeneous graph (users, music, albums, artists) and utilizes the entire user history to form short-term, long-term, and dynamic preferences.

IV. Experiment and analysis of result

We conducted experiments based on the system shown in Figure 1. The experiments were performed with training parameters optimized using the Adam optimizer. In addition, we used the Pytorch library to facilitate model construction.

A. Hyperparameter Tuning

The search for the best parameters is intended for the model to reach its best evaluation point. This process uses 8 random search trials with data size 0.1 and 10 epochs for each parameter combination with the calculation of HR evaluation according to (16) to recommend new music that has never been listened to by users.

The hyperparameter tuning results can be seen in Table I. The 8 random search trials resulted in the HR@10 evaluation in the range of 17 to 22 and the HR@20 in the range of 20 to 26. The best model was achieved with the parameter combination of 0.001 lr, 0.00001 l2, and 100 hidden_size. The combination proves that a large embedding size with sufficient learning rate and small regularization can effectively recognize patterns and thus provide good recommendations.

B. Model Scenarios

One of the influences in achieving good results in our model recommendation system is the influence of preferences. The model scenarios test was conducted with 20 epochs and using the best parameters from the hyperparameter tuning process following Table I.

The results of HR and MRR based on model scenarios can be seen in Table II. Evaluation of the scenario model with next-one resulted in an HR evaluation of 10 in the range of 43 to 44, HR@20 in the range of 47 to 49, and MRR@10 and MRR@20 in the range of 29 to 30. Meanwhile, the next-new recommendation yields an HR@10 evaluation in the range of 32, HR@20 in the range of 36 to 37, and MRR@10 and MRR@20 in the range of 22 to 23.

TABLE I
Results of HR@10 and HR@20 for hyperparameter tuning.

lr	l2	hidden_size	HR@10	HR@20
0.001	0.0001	60	22.13	25.84
0.0005	0.00001	60	17.60	20.90
0.0005	0.001	80	17.64	21.75
0.001	0.00001	80	22.48	25.91
0.001	**0.00001**	**100**	**22.92**	**26.25**
0.0005	0.0001	100	21.72	25.49
0.005	0.0001	100	22.00	24.89
0.0005	0.00001	100	21.35	25.05

TABLE II
Results of HR and MRR based on Model Scenarios.

Next-One Recommendation				
Metric	**Full Preferences**	**W/O Dynamic**	**W/O Long-term**	**W/O Short-term**
HR@10	**44.25**	44.24	43.81	43.14
HR@20	**49.16**	49.06	48.68	47.98
MRR@10	**30.36**	30.24	30.05	29.43
MRR@20	**30.71**	30.58	30.39	29.77
Next-New Recommendation				
Metric	**Full Preferences**	**W/O Dynamic**	**W/O Long-term**	**W/O Short-term**
HR@10	32.69	**32.70**	32.53	32.10
HR@20	**37.06**	37.00	36.79	36.59
MRR@10	**23.34**	23.24	23.29	22.27
MRR@20	**23.64**	23.57	23.59	22.58

The Full Preferences model demonstrates high model performance compared to other scenarios. This indicates that the full integration of the three preference types creates a comprehensive representation of user behavior in the context of music recommendations. This combination enables the model to capture complex patterns in music taste, which often fluctuates based on context, recent user preferences, and long-term taste evolution.

The performance of the W/O Dynamic model is slightly below Full Preferences, but it outperforms it by 0.01 points in next-new recommendations in the HR@10 evaluation, possibly because the narrow scope of recent music preferences limits the diversity of recommendations. This is consistent with music consumption patterns, where the last song listened to is not always a strong predictor of the next song. Conversely, the W/O Long-term model yields the lowest results because it lacks crucial information about the overall listener interaction, failing to connect separate sessions from the same user and unable to form comprehensive preferences.

In the W/O Short-term model, all evaluation metrics are significantly lower than in other scenarios. This significant decline in performance reveals the crucial role of short-term preferences in the context of sequential music recommendations. In music consumption, users tend to listen to songs with similar characteristics or themes in their last session, creating transition patterns that can be captured by the attention mechanism.

C. Comparison with Baseline

For a fair comparison, we conducted a best parameter search test for the baseline model. The best parameters from the

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search test are lr with a value of 0.001, l2 with a value of 0.00001, and hidden_size with a value of 100. After determining the best parameters for the baseline, an evaluation matrix will be calculated to see which model is superior.

The results of the comparison with the baseline are presented in Table III. Our model outperforms the baseline model for all evaluation metrics with an improvement in the range of 1% to 2% for next-one music recommendation and an improvement in the range of 1% for next-new music recommendation.

Table III shows that the performance of the base model that only captures part of the item history is not much better than our model that captures the entire item history. In addition, the model formed from a graph with entity relationships between users, music, albums, and artists performs much better than the graph with no entity relationships between users, music, albums, and artists.

V. Conclusion

In this paper, we implement a combination of GNN with attention mechanisms for sequential music recommendations adapted from the GASM model. GNN capture the relationships between users and items, while attention mechanisms capture changes in preferences during a session. Experiments with Music4All data demonstrate that our model can effectively recommend music despite facing a wide distribution of users. Optimal performance is achieved through appropriate parameters for learning rate, small regularization, and large embedding size. Additionally, the short-term attention mechanism proved to be highly influential, as losing information about changes in user preferences led to a drastic decline in all evaluation metrics. Other preferences, such as long-term and dynamic, also complement each other in creating good recommendations. This was demonstrated when losing one of these preferences caused the model to fail to achieve its best evaluation. In addition, the combination of heterogeneous graphs with user preference and item information enables the model to absorb complex information for accurate recommendations. In the future, research can be conducted by testing the scalability of the model on larger and more diverse music datasets, exploring more varied parameter settings to maximize model evaluation results, and evaluating the model's generalization ability to other recommendation domains such as movies or books to demonstrate the model's capabilities in the SRS approach.

TABLE III
Results of HR and MRR for model comparison with baseline.

Next-One Recommendation				
Model	**HR@10**	**HR@20**	**MRR@10**	**MRR@20**
Baseline (SR-GNN)	43.53	48.39	29.75	30.10
Our Model	44.25	49.16	30.36	30.71
Improvement (%)	1.65	1.61	2.05	2.04
Next-New Recommendation				
Model	**HR@10**	**HR@20**	**MRR@10**	**MRR@20**
Baseline (SR-GNN)	32.34	36.60	22.91	23.20
Our Model	32.69	37.06	23.34	23.64
Improvement (%)	1.10	1.27	1.89	1.89

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